

Literature status: the $\mathcal{B}(\ell^p)$ amenability question

arXiv:math/0401122 answered by arXiv:0907.3984

Status: literature_already_answered, not an original run proof

Source question

The source paper is Narutaka Ozawa, *A note on non-amenability of $\mathcal{B}(\ell_p)$ for $p = 1, 2$* , arXiv:math/0401122, later published in *International Journal of Mathematics* 15 (2004), 557–565. In the introduction, after recalling the known $p = 1, 2, \infty$ cases, Ozawa records the remaining case:

“It is left open whether $\mathcal{B}(\ell_p)$ is amenable or not for $p \in (1, \infty) \setminus \{2\}$.” [1, p. 1]

The queue hit also sees a later remark in Ozawa’s proof discussion: the same strategy would produce almost invariant vectors in $\ell_q \widehat{\otimes} \ell_p$, but Ozawa does not know whether the unit ball of this projective tensor product uniformly embeds into Hilbert space, and notes that nontrivial cotype was not known there. That is a method-side obstruction, not the main amenability question identified above.

Supporting answer

The decisive later paper is Volker Runde, *$\mathcal{B}(\ell^p)$ is never amenable*, arXiv:0907.3984, published in *Journal of the American Mathematical Society* 23 (2010), 1175–1185. Runde explicitly cites Ozawa’s paper and states that Daws and Runde had investigated whether $\mathcal{B}(\ell^p)$ is amenable for $p \in (1, \infty) \setminus \{2\}$ [2, Introduction].

Runde’s final argument gives, for $p \in (1, \infty)$, non-amenability of $\ell^\infty(\mathcal{K}(E))$ for every infinite-dimensional $\mathcal{L}^p$ -space E , then non-amenability of $\mathcal{B}(\ell^p(E))$ for every $\mathcal{L}^p$ -space E [2, Proposition 4.1 and Theorem 4.2]. Applying the corollary with $E = \ell^p$ yields that $\mathcal{B}(\ell^p)$ is not amenable for every $p \in (1, \infty)$ [2, Corollary 4.3 and following paragraph]. Together with the earlier $p = 1, 2, \infty$ results cited there, this answers the open range from Ozawa’s introduction.

Scope and limitations

This packet records a known literature answer to the main open problem in the source introduction. It should not be counted as a new solution, partial solution, or counterexample produced by this run.

No claim is made here that Runde resolves the separate tensor-product uniform-embedding or cotype question mentioned in Ozawa’s remark. Runde instead bypasses that specific obstruction through the Daws–Runde transference route and a modification of Ozawa’s method.

Search and citation evidence

The local run indexes had no existing row for arXiv:math/0401122. Local source inspection found Ozawa’s open statement in the introduction and the separate tensor-product remark near the end of the proof section. A bounded web and local-source search for phrases including `B(e11~p)` amenable, `B(e11~p) is never amenable`, and `Daws Runde Ozawa` found arXiv:0907.3984. The supporting paper’s abstract, introduction, final theorem/corollary, and bibliography identify the answer as the same problem and cite Ozawa’s paper.

References

- [1] N. Ozawa, *A note on non-amenability of $\mathcal{B}(\ell_p)$ for $p = 1, 2$* , arXiv:math/0401122; Internat. J. Math. 15 (2004), 557–565. <https://arxiv.org/abs/math/0401122>
- [2] V. Runde, *$\mathcal{B}(\ell^p)$ is never amenable*, arXiv:0907.3984; J. Amer. Math. Soc. 23 (2010), 1175–1185. <https://arxiv.org/abs/0907.3984>
- [3] M. Daws and V. Runde, *Can $\mathcal{B}(\ell^p)$ ever be amenable?*, Studia Math. 188 (2008), 151–174; erratum Studia Math. 195 (2009), 297–298.